

Week 3

Limits of functions pt 2

Examples

$$\textcircled{1} \lim_{x \rightarrow 3} (x^2 - 2x) = 9 - 6 = 3$$

$$\textcircled{2} \lim_{x \rightarrow a} \frac{x - a}{\frac{1}{x} - \frac{1}{a}}$$

$$= \lim_{x \rightarrow a} \frac{x - a}{\frac{a - x}{ax}}$$

$$= \lim_{x \rightarrow a} (x - a) \cdot \frac{ax}{a - x} = \lim_{x \rightarrow a} \cancel{(x - a)} \cdot \frac{ax}{-\cancel{(x - a)}} = \lim_{x \rightarrow a} ax = -a^2$$

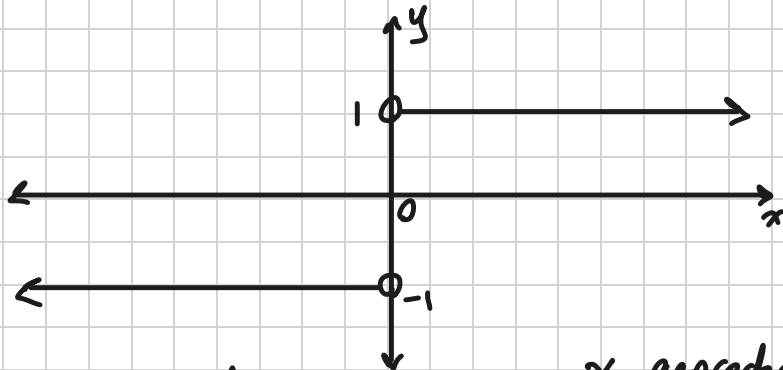
$$\begin{aligned}
 & \textcircled{3} \quad \lim_{t \rightarrow 0} \frac{t}{\sqrt{4+t} - \sqrt{4-t}} \\
 &= \lim_{t \rightarrow 0} \frac{t}{\sqrt{4+t} - \sqrt{4-t}} \cdot \frac{\sqrt{4+t} + \sqrt{4-t}}{\sqrt{4+t} + \sqrt{4-t}} \\
 &= \lim_{t \rightarrow 0} \frac{t(\sqrt{4+t} + \sqrt{4-t})}{4+t - (4-t)} \\
 &= \lim_{t \rightarrow 0} \frac{\cancel{t}(\sqrt{4+t} + \sqrt{4-t})}{2\cancel{t}} \\
 &= \lim_{t \rightarrow 0} \frac{\sqrt{4+t} + \sqrt{4-t}}{2} \\
 &= \frac{\sqrt{4+0} + \sqrt{4-0}}{2} \\
 &= 2
 \end{aligned}$$

One sided limits

Ex

$$f(x) = \begin{cases} -1, & x < 0 \\ 1, & x > 0 \end{cases}$$

$$f(x) = \text{sgn}(x) = \frac{1 \times 1}{x} \quad D(f) = \mathbb{R} - \{0\}$$



Left limit $\lim_{x \rightarrow a^-} f(x)$

x approaches a from the left



$$\lim_{x \rightarrow 0^-} f(x) = -1$$

x approaches a
from the right

Right limit $\lim_{x \rightarrow a^+} f(x)$

$$\lim_{x \rightarrow 0^+} f(x) = 1$$

overall
 $\lim_{x \rightarrow 0} f(x)$ D.N.E

theorem for $\lim_{x \rightarrow a} f(x)$ to exist, the limits $\lim_{x \rightarrow a^-} f(x)$, $\lim_{x \rightarrow a^+} f(x)$ must exist and be equal

Example

$$\textcircled{1} \lim_{x \rightarrow 0} \frac{|3x-1| - |3x+1|}{x}$$

$$|x| = \begin{cases} -x, & \text{if } x < 0 \\ x, & \text{if } x \geq 0 \end{cases}$$

if x is very small, $3x-1 < 0$

$$= \lim_{x \rightarrow 0} \frac{|3x-1| - |3x+1|}{x}$$

$$= \lim_{x \rightarrow 0} \frac{(1-3x) - (3x+1)}{x}$$

$$= \lim_{x \rightarrow 0} \frac{-\cancel{6x}}{\cancel{x}}$$

$$= -6$$

Ex Find $\lim_{x \rightarrow 0^+} \sqrt{x^3 - x}$

① find domain of $f(x) = \sqrt{x^3 - x}$

$\Rightarrow$ require that $x^3 - x \geq 0$

$$x^3 - x = x(x^2 - 1) = x(x-1)(x+1) \geq 0$$

x	$\ominus$	-1	$\ominus$	0	$\oplus$	1	$\ominus$
$x+1$	$\ominus$	0	$\oplus$		$\oplus$		$\oplus$
$x-1$	$\ominus$		$\ominus$		$\ominus$	0	$\oplus$
$x(x^2-1)$	$-$		$+$		$-$		$+$

$$D(f) = [-1, 0] \cup [1, \infty]$$

Small positive values are not in the domain

$$\lim_{x \rightarrow 0^+} \sqrt{x^3 - x} \quad \text{D.N.E}$$

$$\lim_{x \rightarrow 0^-} \sqrt{x^3 - x} = 0$$

this one works
small negative values are in the Domain

$$\lim_{x \rightarrow 0^+} \sqrt[3]{x^3 - x} = 0$$

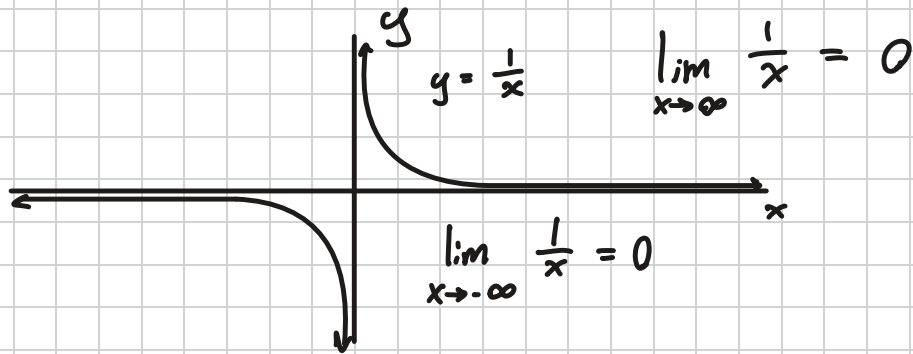
$$\lim_{x \rightarrow 0^-} \sqrt[3]{x^3 - x} = 0$$

} No domain problems

Limits at infinity

$\lim_{x \rightarrow a} f(x)$ a finite

Consider $f(x) = \frac{1}{x}$ $D(f) = \mathbb{R} - \{0\}$

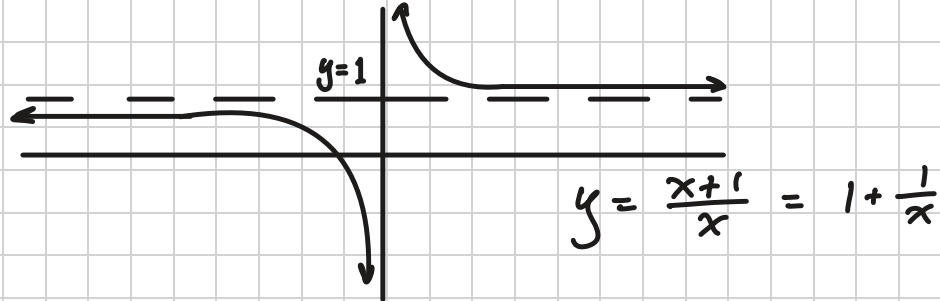


Ex $f(x) = \frac{x+1}{x}$

$$\lim_{x \rightarrow \infty} \frac{x+1}{x}$$
$$= \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)$$

$$= 1$$

Same thing for $\lim_{x \rightarrow -\infty} \frac{x+1}{x} = 1$



Horiz Asymptote @ $y=1$

for $y = \frac{1}{x}$, Asymptote @ $y=0$

Ex $\lim_{x \rightarrow \infty} \frac{1}{\sqrt{x}} = 0$

in general

$$\lim_{x \rightarrow \infty} \frac{1}{x^a} = 0, \quad a > 0$$

Ex

$$\lim_{x \rightarrow \infty} \frac{5x^3 + 3x - 7}{8x^3 + x^2 - 1}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{5x^3 + 3x - 7}{x^3}}{\frac{8x^3 + x^2 - 1}{x^3}}$$

$$= \lim_{x \rightarrow \infty} \frac{5 + \frac{3}{x^2} - \frac{7}{x^3}}{8 + \frac{1}{x} - \frac{1}{x^3}} = \frac{5}{8}$$

(Note: Red circles and arrows in the original image highlight terms that approach 0 as x goes to infinity: 3/x^2, 7/x^3, 1/x, and 1/x^3.)

$$\underline{\underline{Ex}} \quad \lim_{x \rightarrow \infty} \frac{2x+7}{x^2-11}$$

$$= \lim_{x \rightarrow \infty} \frac{\left(\frac{2}{x}\right)^{\rightarrow 0} + \left(\frac{7}{x^2}\right)^{\rightarrow 0}}{1 - \left(\frac{11}{x^2}\right)^{\rightarrow 0}}$$

$$= 0$$

$$\underline{\underline{Ex}} \quad \lim_{x \rightarrow -\infty} \frac{2x-1}{\sqrt{3x^2-x+2}}$$

$$= \lim_{x \rightarrow -\infty} \frac{\frac{2x-1}{x}}{\frac{\sqrt{3x^2-x+2}}{x}}$$

$$= \lim_{x \rightarrow -\infty} \frac{2 - \frac{1}{x}}{-\sqrt{\frac{3x^2-x+2}{x^2}}}$$

$$= \lim_{x \rightarrow -\infty} \frac{2 - \frac{1}{x}}{-\sqrt{3 - \frac{1}{x} + \frac{2}{x^2}}}$$

$$= -\frac{2}{\sqrt{3}}$$

$$\underline{\underline{Ex}} \quad \lim_{x \rightarrow \infty} \frac{2x-1}{\sqrt{3x^2-x+2}} = \frac{2}{\sqrt{3}}$$

Note : $\sqrt{x^2} = |x|$

if $x < 0$

$$\sqrt{x^2} = -x$$

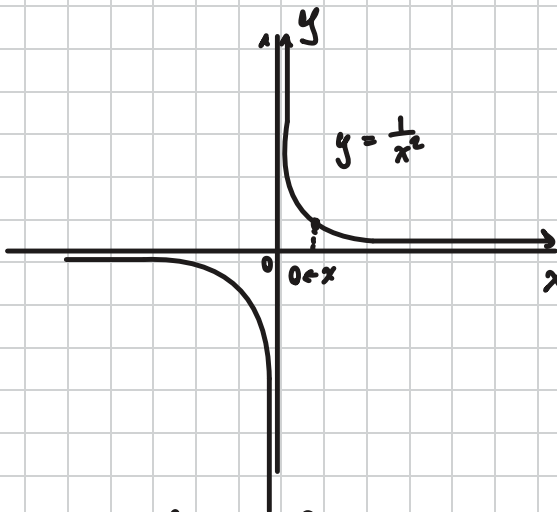
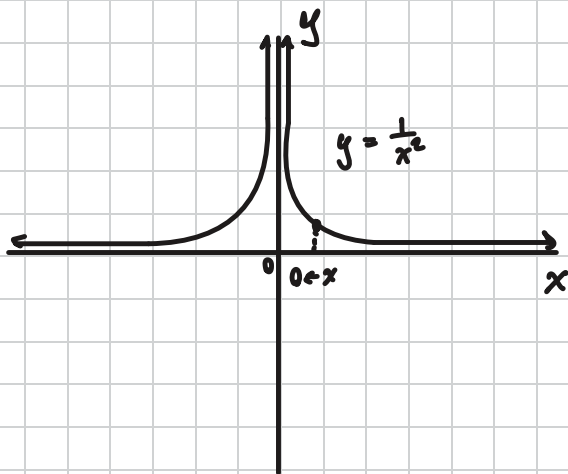
Ex

$$\begin{aligned}
 & \lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x) \\
 &= \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 + x} - x)(\sqrt{x^2 + x} + x)}{\sqrt{x^2 + x} + x} \\
 &= \lim_{x \rightarrow \infty} \frac{\cancel{x^2} + x - \cancel{x^2}}{\sqrt{x^2 + x} + x} \\
 &= \lim_{x \rightarrow \infty} \frac{\frac{x}{x}}{\frac{\sqrt{x^2 + x}}{x} + x} \\
 &= \lim_{x \rightarrow \infty} \frac{1}{\sqrt{\frac{x^2 + x}{x^2}} + 1} \\
 &= \lim_{x \rightarrow \infty} \frac{1}{\sqrt{1 + \frac{1}{x}} + 1} = \frac{1}{2}
 \end{aligned}$$

infinite limits

$$f(x) = \frac{1}{x^2}$$

$$D(f) = \mathbb{R} - \{0\}$$



$$\lim_{x \rightarrow 0^+} \frac{1}{x} = \infty$$

$$\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$$

$$\lim_{x \rightarrow 0} \frac{1}{x} \text{ DNE}$$

(a finite)

$$\text{if } \lim_{x \rightarrow a^+} f(x) = \pm \infty$$

f has a asymptote

Example

$$\lim_{x \rightarrow 0^+} \frac{x+1}{x}$$

$$= \lim_{x \rightarrow 0^+} \left(1 + \frac{1}{x} \right) = \infty$$

$$\lim_{x \rightarrow 0^-} \frac{x+1}{x}$$

$$= \lim_{x \rightarrow 0^-} \left(1 + \frac{1}{x} \right) = -\infty$$

$$\lim_{x \rightarrow \infty} x = \infty, \quad \lim_{x \rightarrow \infty} x^a = \infty, \quad a > 0$$